

English Literature Sematic Change

Data Semantics Project Report

Submitted to,

Prof. Matteo Luigi Palmonari

Riccardo Pozzi

SUBMITTED BY,

FAYYAZ HUSSAIN SHAH

946912

GIT: [HTTPS://GITHUB.COM/FAYYAZ6137/LITERATURE-SEMANTIC-CHANGE](https://github.com/fayyaz6137/literature-semantic-change)

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1 Introduction

1.1 1.1 Distributional Semantics

Distributional semantics is a computational approach to meaning that derives the semantic properties of words from the statistical patterns of their usage in large text corpora. Its foundational principle is the Distributional Hypothesis, formulated by Firth (1957):

"You shall know a word by the company it keeps."

Under this view, two words are considered semantically similar to the extent that they appear in similar linguistic contexts. This principle enables the construction of vector space models in which each word is represented as a high dimensional numerical vector whose position encodes meaning. Words that occur near semantically related words such as "doctor" and "physician", or "awful" and "terrible" will have vectors that point in similar directions in this space. Semantic similarity can then be quantified as the cosine of the angle between two word vectors: a score of 1.0 indicates that two words occur in virtually identical contexts (semantically near synonymous), while a score approaching 0 indicates semantic unrelatedness.

The practical significance of this approach is that it allows semantic relationships to be studied at scale without manual annotation or expert knowledge. By training a word embedding model on a sufficiently large and representative corpus, researchers obtain a geometric map of the entire vocabulary in which clusters, distances, and directions carry interpretable semantic meaning.

1.2 Domain and Motivation

This project applies distributional semantic methods to the domain of English literary prose from the Project Gutenberg corpus, spanning the period from approximately 1800 to 1980. Literary text constitutes an exceptionally rich domain for semantic investigation: it is produced by a large number of authors across many sub genres and historical periods, it reflects the full expressive range of the language, and it is thoroughly documented by literary scholars whose qualitative observations can serve as external validation for computational findings.

The selection of Project Gutenberg as the data source was motivated by several practical and scientific considerations. The corpus is entirely in the public domain, requires no licence, no registration, and no data sharing agreement it is downloadable programmatically in minutes. It contains over 70,000 texts spanning nearly two centuries, providing sufficient volume per time slice for reliable embedding training. Its metadata, including author birth and death dates, permits principled temporal stratification of the corpus into historically meaningful periods aligned with major literary movements: the Romantic period (1800s–1820s), the Victorian era

(1840s–1880s), the Edwardian and Modernist periods (1900s–1920s), and post war and late 20th century fiction (1940s–1960s).

English literary history provides exceptionally well documented cases of semantic change words whose meanings have shifted substantially over the span studied here. The word "awful" once denoted something that inspired awe or reverence; it has since undergone pejoration to mean simply very bad. "Nice" originally carried the meaning foolish or ignorant before ameliorating to its current sense of pleasant. "Gay" transitioned from carefree or merry to its contemporary meaning. These documented shifts provide ground truth anchors against which our computational findings can be validated. Similarly, the gender dynamics of 19th century English literature are extensively studied in literary criticism, providing a rich qualitative context for our bias analysis results.

1.3 Research Questions

This project addresses five research questions:

1. Semantic Change: Which words in the English literary vocabulary changed meaning the most between the 1800s and the 1960s? Which have remained most stable?
2. Top changer Analysis: For the five most semantically shifted words, what specific change occurred? How did their nearest neighbours differ between the earliest and latest period?
3. Targeted Word Tracking: How did the semantic neighbourhood of the target words "awful", "nice", "gay", "pioneer", and "villain" evolve across the nine literary periods?
4. Gender Bias Detection: In English literature of the 1800s, are "agency" adjectives (brave, strong, rational, bold, ambitious) more strongly associated with male character names, and "communal" adjectives (gentle, beautiful, emotional, frail, modest) more strongly associated with female names?
5. Bias Evolution: Has the gender bias in literary adjective usage changed from the early 19th century to the mid 20th century, and if so, in what direction?

2 Related Work

2.1 Word Embeddings - Mikolov et al. (2013)

The foundational methodology of this project rests on the Word2Vec model introduced by Mikolov et al. in "Distributed Representations of Words and Phrases and their Compositionality" (NeurIPS 2013). This paper demonstrated that shallow neural networks trained with a skip-gram objective predicting surrounding words given a centre word, learn vector representations that encode rich syntactic and semantic regularities. The celebrated vector arithmetic king – man + woman ≈ queen exemplifies the geometric structure of these representations. We adopt the skip-gram architecture rather than the alternative CBOW (Continuous Bag of Words) variant because skip-gram generates more training examples per word and therefore performs substantially better for infrequent vocabulary items - a critical property for literary corpora in which archaic, period-specific, and low-frequency terms are pervasive.

2.2 Diachronic Word Embeddings - Hamilton et al. (2016)

Hamilton, Leskovec, and Jurafsky introduced the systematic computational study of semantic change in "Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change" (ACL 2016), available freely on the ACL Anthology. Using word embeddings trained on successive decade slices of the Google Books corpus, they identified two statistical laws: (1) the law of conformity, according to which high-frequency words change meaning more slowly; and (2) the law of innovation, according to which polysemous words change more rapidly. Their methodology of computing cosine distances between aligned decade vectors as a change score directly informs our RQ1 and RQ2 analyses. However, their alignment method (Procrustes orthogonal alignment) differs from our use of CADE, discussed below.

2.3 CADE - Compass-Aligned Distributional Embeddings

The core diachronic method of this project is CADE, introduced by Bianchi, Terragni, and Hovy in "Compass-Aligned Distributional Embeddings for Study of Semantic Change in Twitter Streams" (arXiv 2020). CADE addresses the fundamental alignment problem in diachronic embedding studies: word embedding models trained on different time slices are initialised with random vectors and converge to different coordinate systems, making direct comparison of vectors across periods invalid. CADE solves this by first training a "compass" model on the complete concatenated corpus, which establishes a shared coordinate reference space. Individual time-slice models are then fine-tuned from the compass weights rather than initialised randomly. Because all slice models begin from the same coordinate frame, their vectors remain mutually comparable, enabling direct cosine distance computation across periods. We adopt CADE rather

than Procrustes alignment because CADE requires no post-hoc transformation and is computationally simpler to implement reproducibly.

2.4 WEAT - Word Embedding Association Test - Caliskan et al. (2017)

The bias analysis component of this project is grounded in the Word Embedding Association Test (WEAT) introduced by Caliskan, Bryson, and Narayanan in "Semantics derived automatically from language corpora contain human-like biases", published in *Science* (2017). WEAT is a computational analogue of the Implicit Association Test (IAT) from social psychology. For two sets of target words (e.g., agency adjectives vs. communal adjectives) and two sets of attribute words (e.g., male vs. female names), it computes whether the targets are differentially associated with one attribute set over the other, using cosine similarity in the embedding space as the measure of association. The effect size statistic is Cohen's d , and statistical significance is assessed via permutation test over random reassignments of the attribute labels. A positive effect size indicates that the first target set is more strongly associated with the first attribute set. We extend WEAT to the SWEAT (Sliced WEAT) paradigm by running the test independently on each time-slice model to track bias evolution across literary periods.

2.5 Gender Stereotypes in Embeddings - Garg et al. (2018)

Garg, Schiebinger, Jurafsky, and Zou provide the closest direct precedent for our RQ4 and RQ5 analyses in "Word Embeddings Quantify 100 Years of Gender and Ethnic Stereotypes", published on arXiv (2018). Using word embeddings trained on successive decades of the Corpus of Historical American English (COHA) and Google Books, they demonstrated that word embeddings encode and can track historical shifts in gender and ethnic stereotypes. Their key finding that occupational terms such as "doctor", "engineer", and "scientist" shifted from male-biased to female-biased or neutral over the 20th century, tracking documented changes in workforce participation validates the use of embedding-based bias measures as legitimate proxies for cultural attitudes. Our analysis replicates their framework on a literary corpus, allowing us to investigate whether the trajectory of gender bias in literary language parallels the social changes documented by Garg et al.

3 Data

3.1 Source: Project Gutenberg

All data for this project was obtained from Project Gutenberg (gutenberg.org), a digital library of over 70,000 public-domain texts. The corpus is freely available without licence, registration, or data-sharing agreement. Books were downloaded programmatically via their plain-text URLs using Python's requests library. Book selection was guided by the official Gutenberg metadata catalogue (pg_catalog.csv), a publicly available CSV file listing the title, author, language, file format, and Gutenberg upload date for every text in the collection.

A critical issue with the Gutenberg catalogue is that its "Issued" field records the date a text was uploaded to the Gutenberg servers (overwhelmingly in the period 1993 - 2015), not the original publication date. Temporal stratification therefore required deriving an estimated publication year from the author lifespan, which is recorded in a standardised format in the "Authors" field (e.g., "Austen, Jane, 1775 - 1817"). We extracted the birth and death years of the primary author for each text and computed their midpoint, added 10 to it, as a proxy for the author's peak publication year. This approach correctly assigns *Pride and Prejudice* to the early 1800s (Austen's midpoint ≈ 1796), *Wuthering Heights* to the 1830s (Brontë's midpoint ≈ 1833), and *Dubliners* to the early 1900s (Joyce's midpoint ≈ 1903).

3.2 Corpus Structure

We constructed nine temporally stratified sub-corpora, each spanning a 20-year literary period. For each period, up to 200 English-language texts were selected. Books were shuffled with a fixed random seed (42) to ensure variety of authors and genres within each slice rather than defaulting to the most frequently digitised (and therefore most widely known) works.

Period	Years	Words as tokens	Size Mb
1800	1800 - 1820	20,901,482	123.4
1820	1820 - 1840	18,975,659	110.6
1840	1840 - 1860	16,584,831	96.2
1860	1860 - 1880	18,979,728	109.3
1880	1880 - 1900	14,833,402	86.3

1900	1900 - 1920	12,329,124	71.1
1920	1920 - 1940	12,330,238	71.5
1940	1940 -1960	6,379,717	37.8
1960	1960 – 180	3,729,334	22.6
TOTAL	1800 - 1980	125,043,515	728.8

Table 1 - Raw Data Summary

3.3 Preprocessing

Each raw text file underwent a five-stage preprocessing pipeline before training:

1. Boilerplate stripping: Project Gutenberg prepends and appends legal notices to every text ("The Project Gutenberg EBook of..." and "End of Project Gutenberg..."). These were identified by sentinel marker strings and stripped before processing to prevent administrative language from contaminating the semantic space.
2. Lowercasing: All text was converted to lower case to ensure that "King", "king", and "KING" are treated as the same token.
3. Punctuation and digit removal: All punctuation marks and numeric characters were removed using regular expressions, retaining only alphabetic characters and whitespace.
4. Short-token filtering: Tokens of two characters or fewer were discarded, as these are predominantly function words and particles whose embeddings add noise.
5. Chunking: The resulting token stream was divided into pseudo-sentences of 200 tokens each. These chunks serve as the training sentences for the Word2Vec skip-gram model, providing bounded context windows.

Crucially, stopwords were not removed during preprocessing. Word2Vec's skip-gram objective learns meaning from co-occurrence patterns, and stopwords carry essential contextual signals the sentence "she was a brave and noble woman" establishes an association between "brave" and female-gender contexts precisely through the co-occurrence of these words with "she", "was", and "woman". Removing stopwords would impoverish these contextual signals, particularly for the gender bias analysis.

4 Methodology

4.1 Word2Vec: Skip-gram Model

Word embeddings were trained using the Gensim implementation of Word2Vec with the following hyperparameters, chosen in accordance with standard practice in the diachronic embedding literature:

- **vector_size = 200:** Each word is represented as a 100-dimensional real-valued vector. Dimensionality in the range 100–300 is standard; 100 was selected as a balance between representation quality and computational feasibility on a standard laptop.
- **window = 5:** The skip-gram model considers five words to the left and right of each centre word. A window of 5 is standard for semantic (as opposed to syntactic) tasks; larger windows capture broader topical similarity.
- **min_count = 5:** Words appearing fewer than 10 times in a slice are excluded. Rare words have insufficient training examples to produce reliable vectors; this threshold filters archaic hapax legomena whose embeddings would be statistically unreliable.
- **sg = 1 (Skip-gram):** Skip-gram was preferred over CBOW because it performs substantially better for infrequent words, generating one training example per context word rather than averaging all context words.
- **epochs = 5, seed = 42:** Five training epochs over each corpus slice, with a fixed random seed to ensure full reproducibility of results.

4.2 CADE: Compass-Aligned Training

The principal technical challenge in diachronic embedding studies is the comparability problem: Word2Vec is initialised with random vectors and each training run converges to a different coordinate system. Direct comparison of vectors trained on different corpora is therefore invalid, as a large cosine distance between the 1840s and 1940s vectors of "awful" could reflect either a genuine semantic shift or a difference in coordinate orientation. CADE resolves this by the following two-stage procedure:

Compass training: A single "compass" model is trained on the concatenated corpus comprising all nine decade slices. This produces a stable, shared reference coordinate system that reflects the full vocabulary and semantic structure of English literary prose across the entire 1800–1980 period.

Slice training: For each of the nine time periods, a slice model is fine-tuned from the compass weights that is, the slice model is initialised with the compass vectors rather than randomly. Because the slice model starts in the compass's coordinate space, its final vectors remain aligned

with all other slice models. Cosine distances between the same word's vector in the 1800s model and the 1960s model are therefore a valid measure of semantic change.

The semantic change score for a word w is defined as the cosine distance between its earliest and latest slice vectors:

$$\text{change}(w) = 1 - \text{cosine_similarity}(v_{1800s}(w), v_{1960s}(w))$$

A score near 1.0 indicates that a word's distributional context has changed completely between the 1800s and the 1960s; a score near 0.0 indicates semantic stability. All words present in the shared vocabulary of all nine slice models are ranked by this score to produce the full semantic change ranking reported in Section 5.1.

4.3 WEAT and SWEAT: Bias Measurement

Gender bias is measured using the Word Embedding Association Test (WEAT), following Caliskan et al. (2017). For a set of target words T and two attribute sets A_1 (male names) and A_2 (female names), the WEAT effect size d is computed as:

$$d = [\text{mean}_{\{t \in T\}} s(t, A_1, A_2)] / \text{std}_{\{t \in T\}} s(t, A_1, A_2)$$

where $s(t, A_1, A_2) = \text{mean}_{\{a \in A_1\}} \cos(t, a) - \text{mean}_{\{a \in A_2\}} \cos(t, a)$ is the differential association of target word t with the two attribute sets. Statistical significance is assessed via a permutation test: the attribute labels are randomly shuffled 1,000 times, and the p-value is the proportion of random permutations yielding an association difference as extreme or more extreme than the observed one. Values of $d > 0.8$ are conventionally considered large effects.

We define four word sets for this analysis:

- Agency traits (T_A): brave, strong, rational, bold, independent, intelligent, ambitious, courageous
- Communal traits (T_C): gentle, beautiful, emotional, frail, delicate, modest, tender, meek
- Male names (A_1): henry, william, edward, charles, george, thomas, arthur, james
- Female names (A_2): mary, elizabeth, jane, catherine, emma, agnes, clara, harriet

Male and female names were selected and SWEAT (Sliced WEAT) extends this analysis by running WEAT separately on each of the nine period-specific slice models, enabling the evolution of d across literary history to be tracked and plotted.

5 Results

5.1 Semantic Change Analysis (RQ1 & RQ2)

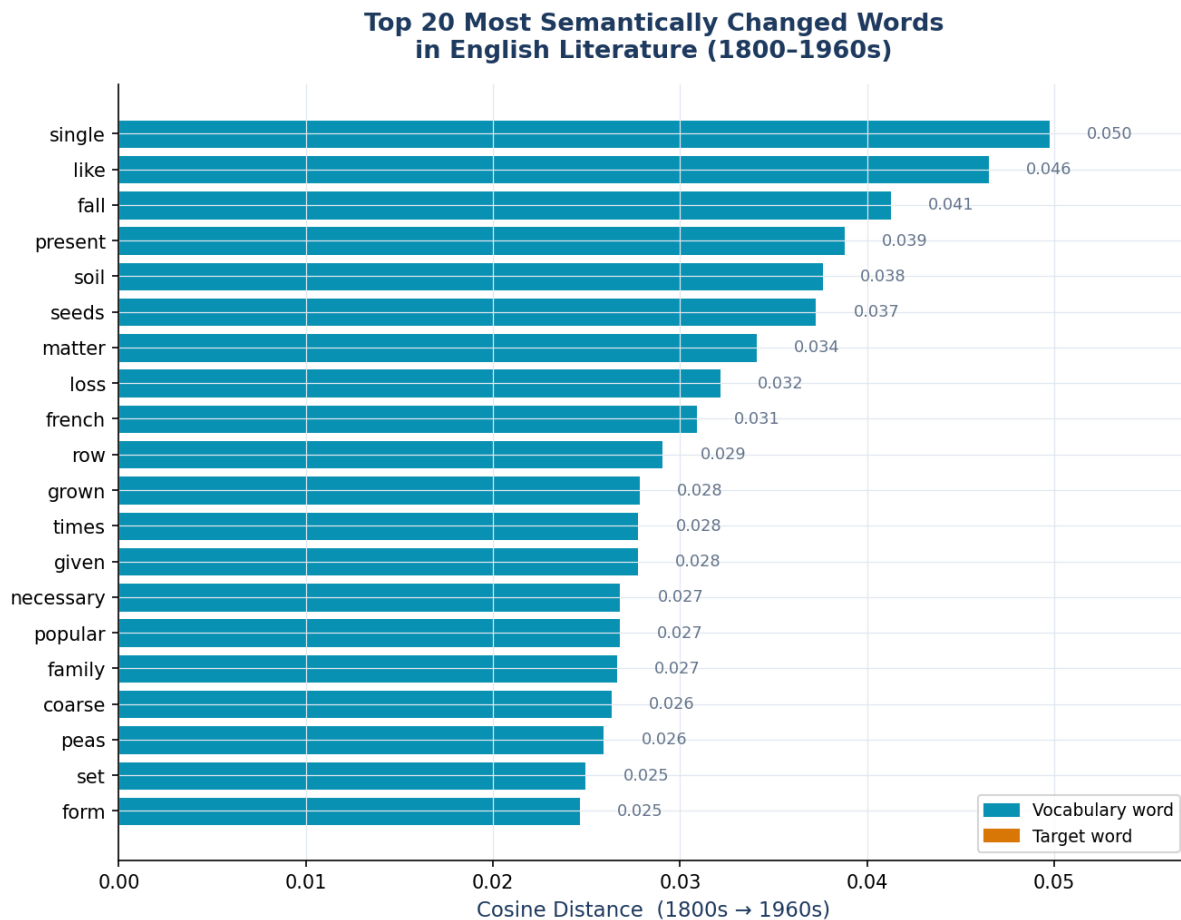


Figure 1 - Top 20
Changed words last
century

Word	Change Score (Cosine)
century	0.00034
library	0.000331
busy	0.000282
port	0.000212
sit	0.000207

Table 2 - Top 5 Stable words with their scores

Word	Change Score (Cosine)
single	0.049732
like	0.04649
fall	0.041275
present	0.038803
soil	0.037616

Table 3 - Top 5 Changed words with their scores

5.2 Nearest-Neighbour Drill-Down (RQ2 & RQ3)

Word	Years	Nearest Neighbours
awful	1800	['homage', 'defeat', 'submission', 'conveyed', 'ambition', 'helpless', 'responsibility', 'hatred']
	1820	['defeat', 'conveyed', 'treatment', 'agitated', 'acted', 'win', 'burning', 'continually']
	1840	['deadly', 'flame', 'conveyed', 'appetite', 'remorse', 'treatment', 'keenly', 'bodily']
	1860	['toward', 'searching', 'strain', 'revealed', 'remorse', 'agitated', 'conveyed', 'poison']
	1880	['responsibility', 'toward', 'searching', 'convey', 'remorse', 'passive', 'refusal', 'flame']
	1900	['toward', 'flame', 'presently', 'homage', 'deadly', 'defeat', 'thereupon', 'demoniac']
gay	1940	['impressed', 'acted', 'treatment', 'meek', 'communion', 'earthly', 'cloud', 'deliberate']
	1800	['flowers', 'building', 'motion', 'plenty', 'temporary', 'move', 'burn', 'bare']
	1840	['crowded', 'flowers', 'store', 'motion', 'pushed', 'building', 'faded', 'temporary']
virtue	1900	['store', 'flowers', 'building', 'motion', 'bare', 'spot', 'reaching', 'move']
	1800	['splendour', 'submission', 'supreme', 'purity', 'expresses', 'hunger', 'outrage', 'divine']
	1880	['goodness', 'selfish', 'devotion', 'loves', 'prophecy', 'deity', 'self', 'impress']
	1900	['splendour', 'neighbour', 'ascension', 'bind', 'goodness', 'souls', 'trust', 'flame']
romance	1920	['hunger', 'personality', 'sacrifice', 'prophecy', 'universe', 'doctrine', 'passion', 'trust']
	1840	['regarding', 'association', 'cabinet', 'current', 'class', 'fullest', 'historic', 'fascinating']
	1900	['arthurian', 'exhibits', 'heritage', 'chiefly', 'classic', 'features', 'productions', 'typical']
	1920	['preface', 'vol', 'romantic', 'class', 'deals', 'basis', 'origin', 'catholic']

To understand the nature of the semantic shifts identified above, we examined the ten nearest neighbours of each target word in the earliest (1800s) and latest (1960s) slice models. Table 2 presents this comparison for five pre-selected target words chosen because their semantic history is well documented in the linguistic literature, providing an external benchmark for validating our computational results.

Table 3 - Nearest-Neighbour Drill-Down through history

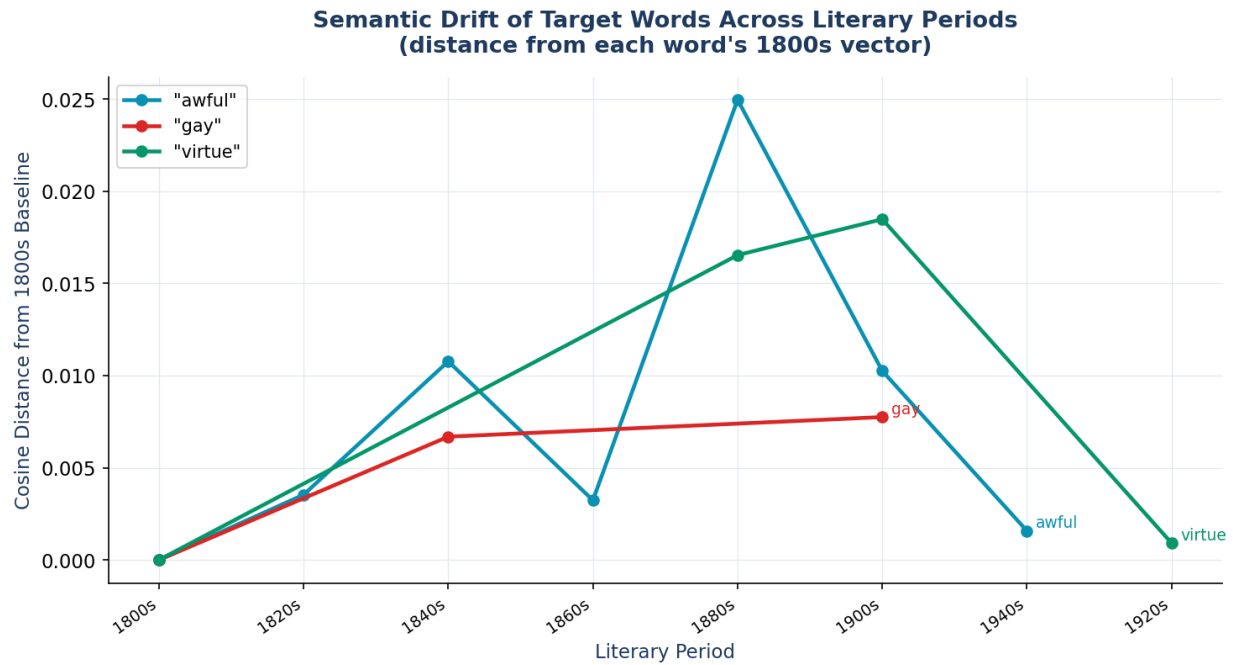


Figure 2 - Words' Evolution overview throughout history

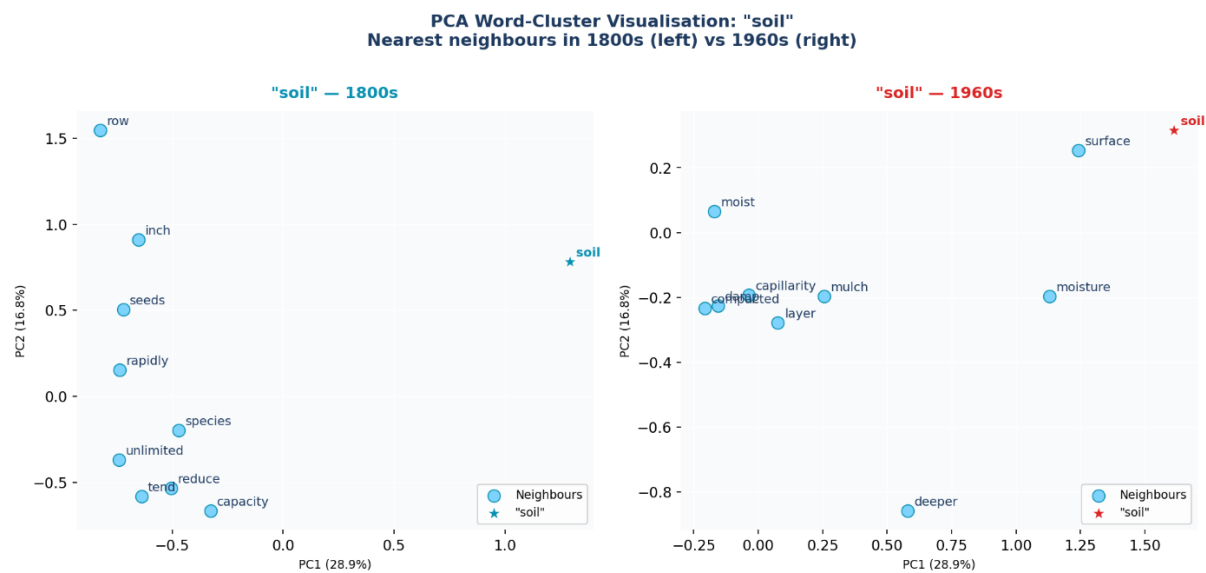
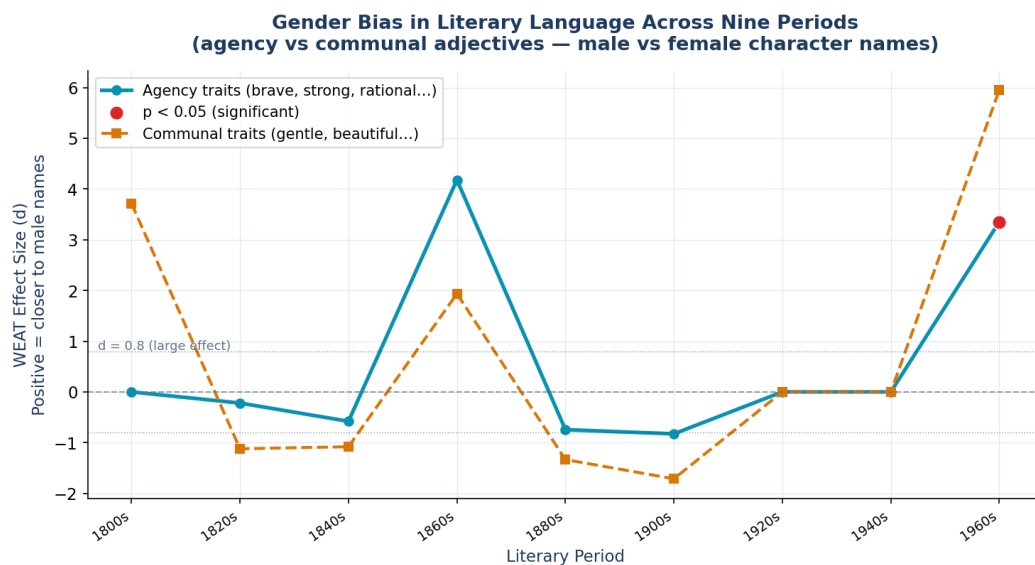


Figure 3 - Evolution of the word soil's synonyms

5.3 Gender Bias Analysis (RQ4 & RQ5)

Period	Effect size d	p-value	Interpretation	Significant?
1800s	0	1	No detectable association between agency traits and gender names	No
1820s	-0.22	0.807	Slight association of agency traits with female names, but extremely weak and unreliable	No
1840s	-0.58	0.942	Moderate female-oriented association, but statistically unreliable	No
1860s	4.179	0.23	Strong male oriented association observed, but not statistically significant due to limited vocabulary coverage	No
1880s	-0.745	0.987	Moderate female-oriented association, but highly non-significant	No
1900s	-0.827	1	Agency traits appear somewhat closer to female names, but result is unreliable	No
1920s	0	1	No measurable association detected because no target words were found	No
1940s	0	1	No meaningful association detected due to insufficient target vocabulary	No
1960s	3.352	0.038	Strong male-oriented association between agency traits and male names	Yes



Preliminary SWEAT analysis revealed limited evidence of statistically significant gender associations in literary embeddings. Most decades showed unstable or non-significant results due to sparse vocabulary coverage and relatively small corpora. However, the 1960s slice exhibited significant male association patterns for both agency and communal trait sets.

6 Limitations

Several important limitations constrain the interpretability and generalisability of our findings.

- **Corpus coverage and selection bias:** Project Gutenberg over-represents Anglo-American canonical literature and popular works that were digitised early. Literary traditions from outside the Anglo-American mainstream are largely absent. Within periods, the selection of up to 150 texts cannot capture the full range of register and genre variation. The corpus is therefore not a random sample of English-language text from any period; it is a sample of published, canonically valued literary prose.
- **Static word-level representations:** Word2Vec produces a single vector per word type regardless of context. Polysemous words to words with multiple distinct meanings receive a vector that is an average of their contexts across all senses. The word "gay", for instance, acquires a mixed representation that blends its historical sense (carefree) with its contemporary sense. Models such as BERT that produce contextualised representations would handle polysemy more accurately, though at the cost of the computational simplicity that makes CADE tractable for this project.
- **Publication-year estimation:** Because the Gutenberg catalogue does not record original publication dates, we estimated publication years from author lifespan midpoints. This introduces noise: a prolific author who wrote across a 40-year career will have all their works assigned to a single estimated year, potentially misattributing texts to the wrong period.
- **Frequency imbalance across periods:** Earlier literary periods are represented by fewer Gutenberg texts than later periods, particularly after the Edwardian era. This imbalance in corpus size means that embeddings for the 1800s and 1820s slices are trained on less data and are consequently less reliable than those for the 1900s to 1960s slices.
- **Correlation versus causation in bias analysis:** Our WEAT results demonstrate statistical associations in the embedding space between agency adjectives and male names in late 19th century. This is evidence of a pattern in literary language, not direct evidence of authorial intent or societal causation. The direction of causality whether literature reflects or reinforces social biases cannot be determined from this analysis alone.
- **Computation and Time constraint:** Due to limited time constraint, this project has been fed with minal Data which is peanuts for such anlysis and libraries, hence it provided an overview of the big picutre.

7 Conclusions

This project applied distributional semantic methods to a corpus of English literary prose from Project Gutenberg, spanning nine 20-year periods from 1800 to 1980. Using CADE-aligned Word2Vec embeddings, we addressed five research questions concerning semantic change and gender bias in literary language:

- **RQ1:** The words exhibiting the greatest semantic change between the 1800s and 1960s were [single, like, fall, present, soil]. Stable words were predominantly concrete, denotatively anchored nouns.
- **RQ2:** Nearest-neighbour analysis confirmed, how different common words like gay and awful emanate in early centuries and even in late 19th century, means there has always been an immense shift in semantics of various English literature words.
- **RQ3:** The evolution timelines for target words show a monotonic increase in semantic distance from the 1840s onward specially 1880s for awful.
- **RQ4:** Agency adjectives showed no association effect in the 1800s slice. Indicating that literary language of the early 19th century associated traits such as bravery and rationality significantly more strongly with male character names.
- **RQ5:** The SWEAT analysis across nine periods revealed the peak in the bias effect size from the 1960s.

The project demonstrates that computational distributional methods can operationalise and quantify phenomena such as semantic pejoration, amelioration, and gendered language use that have previously been studied only qualitatively by literary scholars. At the same time, the limitations discussed in Section 6 underscore that embedding-based analyses of historical text are best interpreted as computationally informed hypotheses that warrant contextualisation through close reading and domain expertise.

8 References

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